

SYNTHESIS, DEGRADATION OF HEMOGLOBIN AND IRON METABOLISM

Dr. Fatima Haq

Lecturer

DIKIOHS, DUHS.

LEARNING OBJECTIVES

At the end of this lecture students should be able to:

- Describe the structure and function of haemoglobin
- Explain the steps involved in haemoglobin synthesis
- Identify the sites involved in haemoglobin synthesis
- Correlate abnormalities in haemoglobin synthesis
- Outline the process of haemoglobin degradation and the formation of bilirubin
- Describe the transport, storage and regulation of iron in the body
- Explain the role of certain proteins like ferritin and transferrin in iron metabolism

INTRODUCTION

Hemoglobin(Hb) is the iron containing colouring matter of red blood cells(RBCs). Almost all vertebrates contain haemoglobin with the sole exception of fish family *channichthyidae* in which most ice-fish have lost their haemoglobin as an adaptation to cold water. Haemoglobin is a *chromoprotein* forming 95% of dry weight(without water) of RBC & 30-34% of wet weight(with water) of RBC. Molecular weight of normal adult haemoglobin(Hb A) is 64,458.

HAEMOGLOBIN CONTENT IN BLOOD

Average haemoglobin content in blood is 14 to 16 g/dl. However, the value of haemoglobin varies depending upon the age & sex of individual. At the time of birth, Hb content is very high because of increased no. of RBCs.

At birth: 25g/dl

After 3rd month: 20 gm/dl

After one year: 17 gm/dl

From puberty onwards: 14-16 gm/dl

Normal Hb content in adult males is 15 gm/dl while in adult females is 14.5 gm/dl.

FUNCTIONS OF HAEMOGLOBIN

Haemoglobin performs two important functions in the body:

- 1) Transport of respiratory gases(oxygen & carbondioxide).
- 2) Buffer activity.

•TRANSPORT OF GASES

The main function of haemoglobin is to transport respiratory gases i.e. transport oxygen from lungs to tissues & carbondioxide from tissues to lungs.

Transport of oxygen: When oxygen binds with haemoglobin, oxygenation process takes place and **oxyhaemoglobin** is formed. Oxyhaemoglobin is an unstable compound and the combination is reversible i.e. when more oxygen is available it combines with haemoglobin and whenever oxygen is required, haemoglobin can release oxygen readily. When oxygen is released from haemoglobin, it is called **reduced haemoglobin** or **ferrohaemoglobin**

CONT..

Transport of carbondioxide: When carbondioxide binds with haemoglobin, carbhaemoglobin is formed. It is also an unstable compound and the combination is reversible i.e. the carbondioxide can be released from this compound.

The affinity of haemoglobin for carbondioxide is 20 times more than that for oxygen.

- **BUFFER ACTION**

Haemoglobin acts as a buffer and plays an important role in maintaining the acid-base balance of the body.

STRUCTURE OF HAEMOGLOBIN

Haemoglobin is a conjugated globular protein. It consists of protein combined with an iron-containing pigment. The iron-containing pigment is ***heme*** and the protein part is ***globin***. Heme also forms a part of structure of myoglobin(oxygen-binding pigment in muscles) & neuroglobin(oxygen-binding protein in brain).

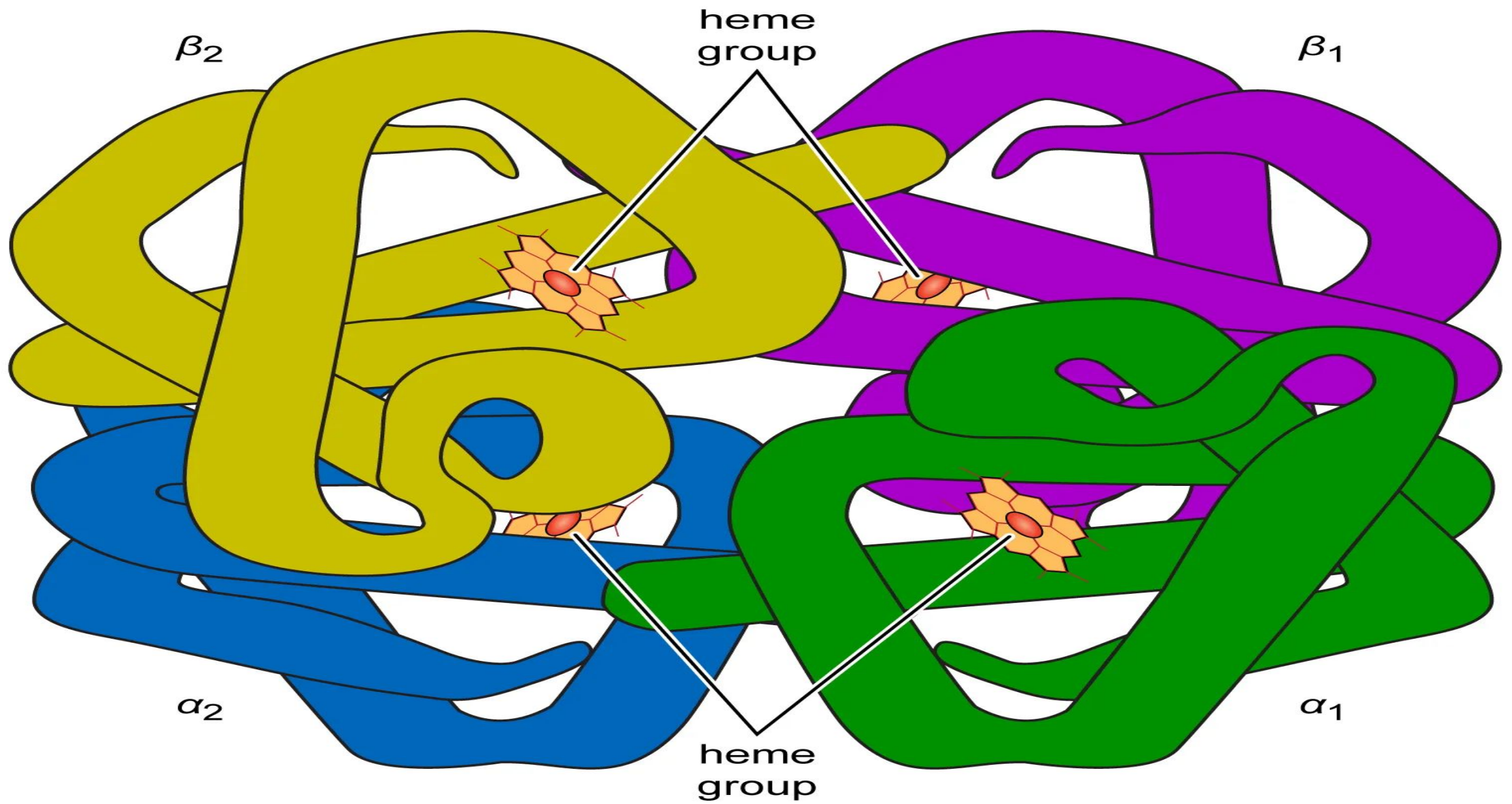
Haemoglobin has a quaternary structure with two pairs of polypeptide chains(heterotetramer) called ***alpha*** and ***beta*** chains.

ADULT HEMOGLOBIN(Hb A)

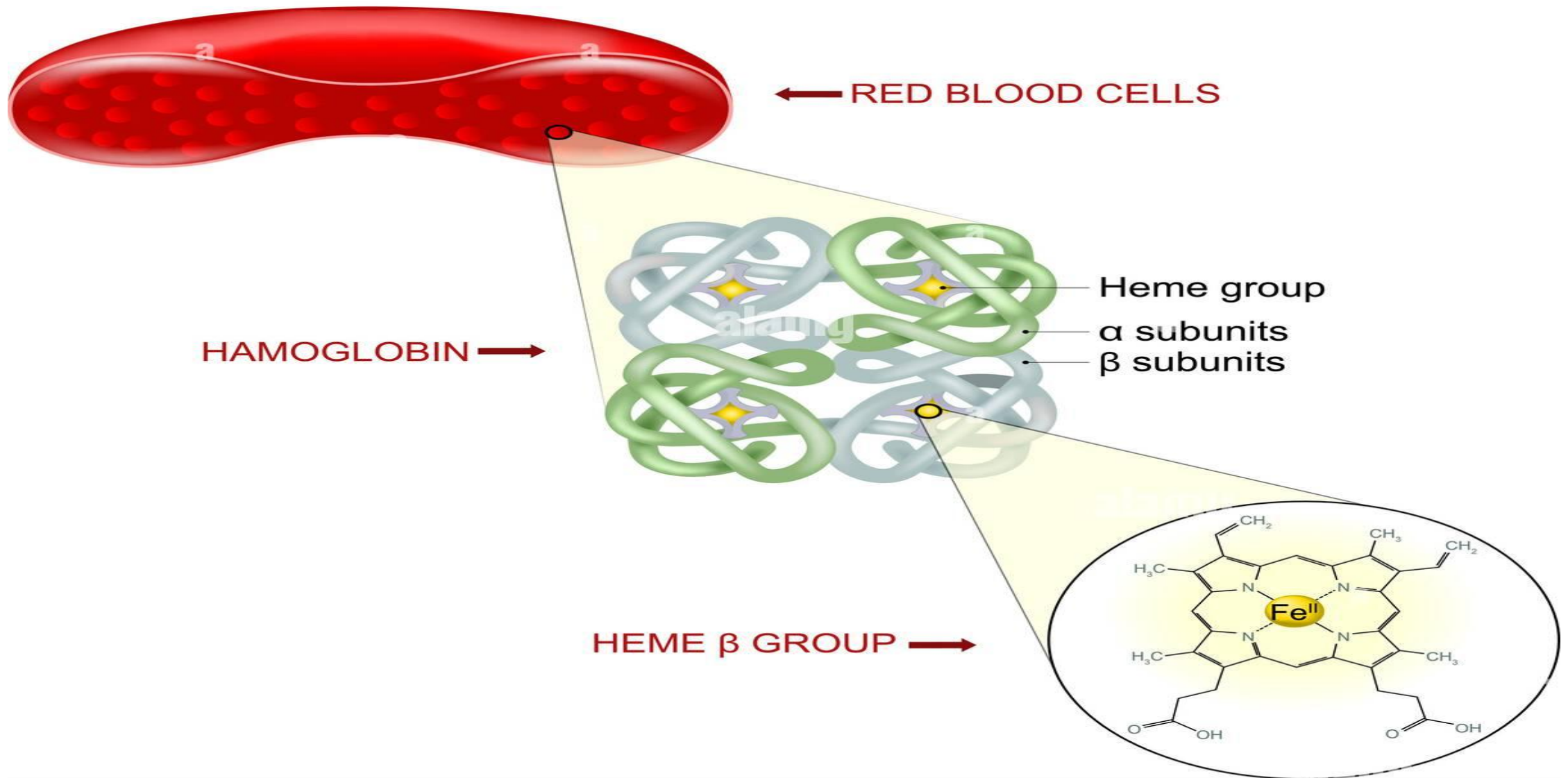
Adult hemoglobin has two alpha chains(each of 141 amino-acids) & two beta chains(each of 146 amino-acids).

DIAMERS

Alpha & beta dimers as $\alpha_1\beta_1$ & $\alpha_2\beta_2$ linked by weak ionic bond.



Structure of human hemoglobin



HAEMOGLOBIN FORMATION

Haemoglobin formation(Hb synthesis) is a complex process occurring in developmental red blood cells within the bone marrow. The synthesis of haemoglobin begins in polychromatophil erythroblasts and continues even into the reticulocyte stage of RBCs formation. Therefore when reticulocyte leaves the bone marrow and pass into the blood stream, they continue to form minute quantities of haemoglobin for another day or so until they become mature erythrocytes.

Haemoglobin consists of two parts **heme** and **globin**. Heme synthesis takes place in mitochondria and cytosol of immature red blood cells. In the formation of heme part of haemoglobin, **succinyl coA** which is formed in the krebs metabolic cycle binds with **glycine** to form a **pyrrole** molecule. Four pyrrole molecules combine to form **protoporphyrin IX**.

Cont..

Protoporphyrin IX then combines with iron to form the heme molecule. Finally each heme molecule combines with a long polypeptide chain a globin. Globin is synthesized by ribosomes forming a subunit of haemoglobin. Each chain has a molecular weight of about 16000. Four of these chains in turn bind together loosely to form the whole haemoglobin molecule. Because each haemoglobin chain has a heme prosthetic group containing an atom of iron, and there are four haemoglobin chains in each haemoglobin molecule so there are total four iron atoms in each haemoglobin molecule. Each of these atoms of iron can bind loosely with one molecule of oxygen(or eight oxygen atoms) that can be transported by each haemoglobin molecule.

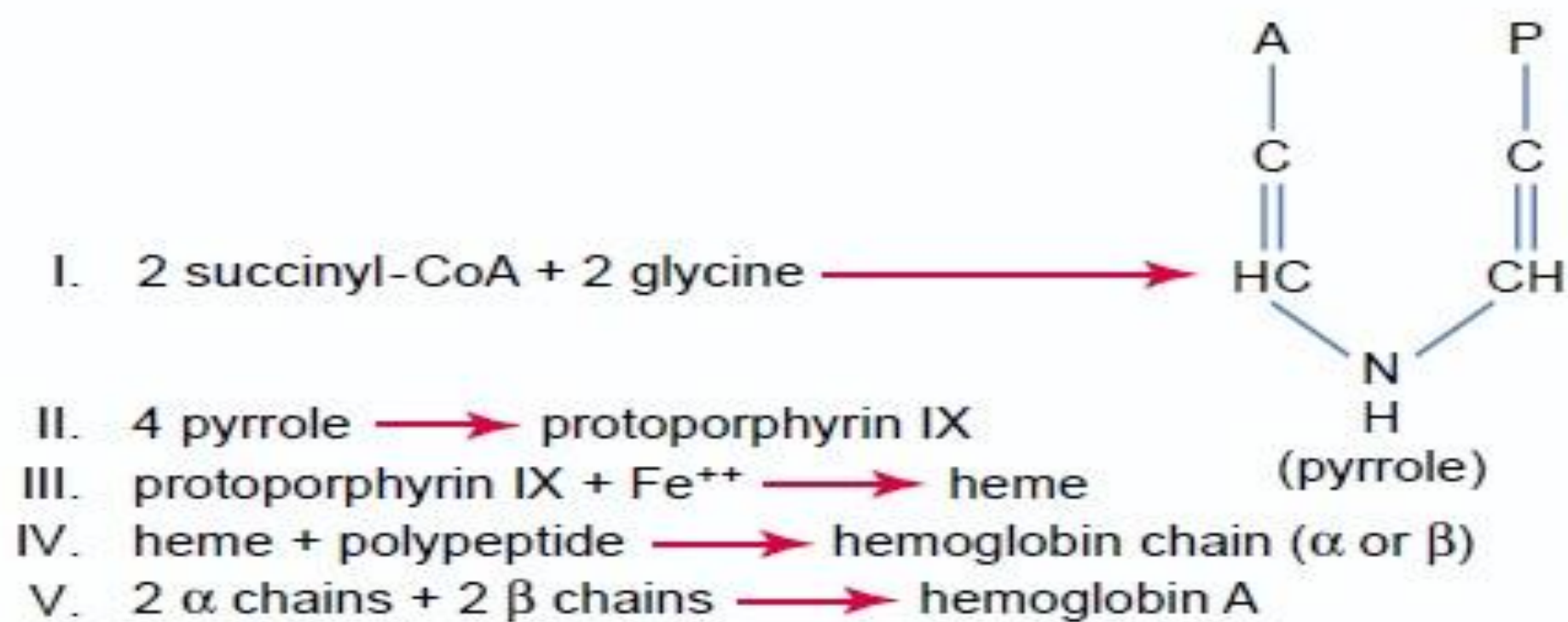
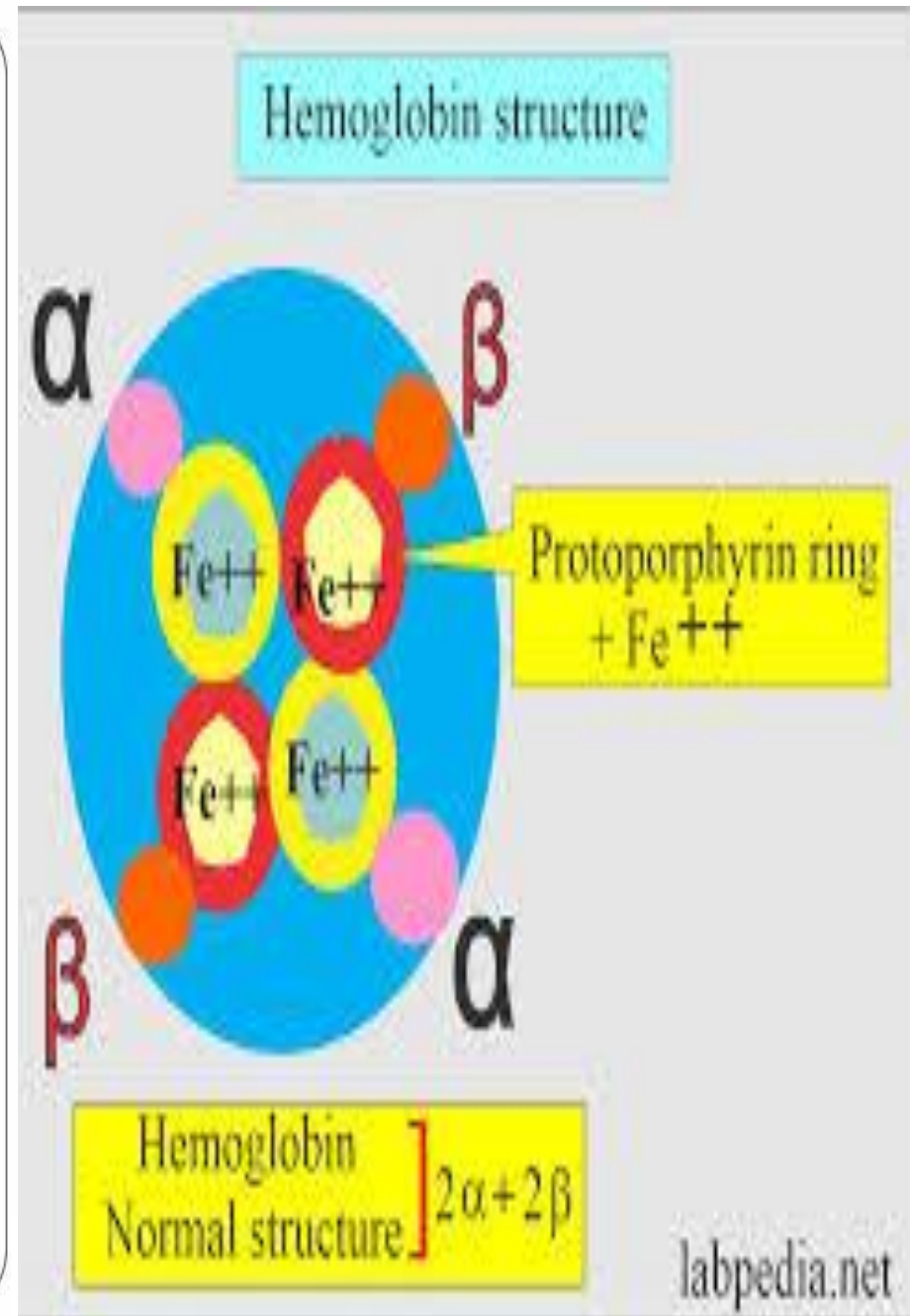


Figure 32-5

Formation of hemoglobin.

HAEMOGLOBIN SYNTHESIS

- Each molecule of haem combines with a globin chain.
- A tetramer of four globin chains
- each with its own haem group
- formed to make up a haemoglobin molecule



VARIATIONS IN POLYPEPTIDE CHAINS

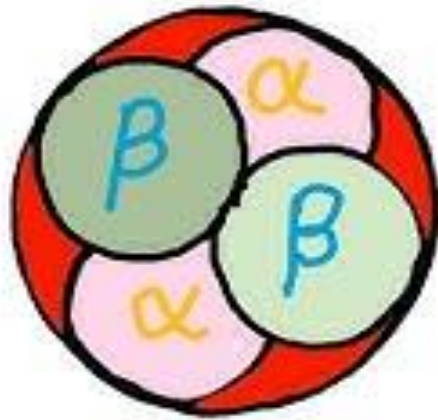
There are several slight variations in the different subunit haemoglobin chains depending upon the amino-acid composition of polypeptide chains.

These variations are designated as:

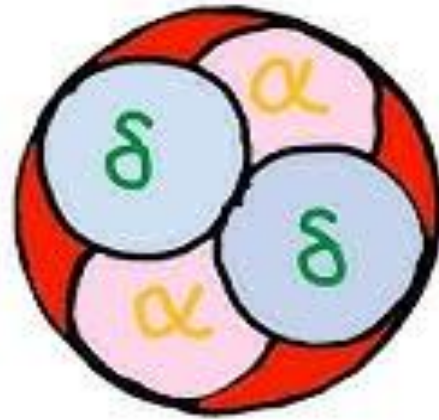
- Alpha chains
- Beta chains
- Gamma chains
- Delta chains

The types of haemoglobin chains in the haemoglobin molecule determine the binding affinity of haemoglobin for oxygen. The most common form of haemoglobin in adults (*haemoglobin A*) is a combination of *two alpha chains* and *two beta chains*. Haemoglobin A has a molecular weight of 64,458.

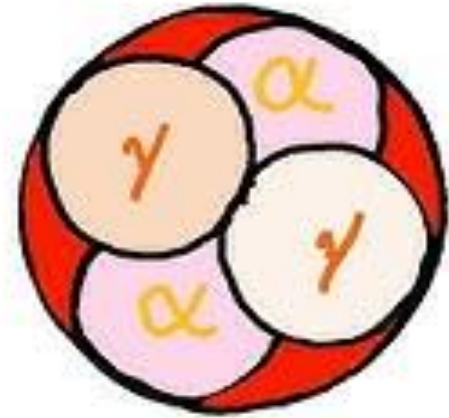
Types of Hemoglobin



HbA



HbA₂



HbF

OXYGEN-HEMOGLOBIN DISSOCIATION CURVE

GRAPH PLOT:

Y-axis = % saturation of hemoglobin

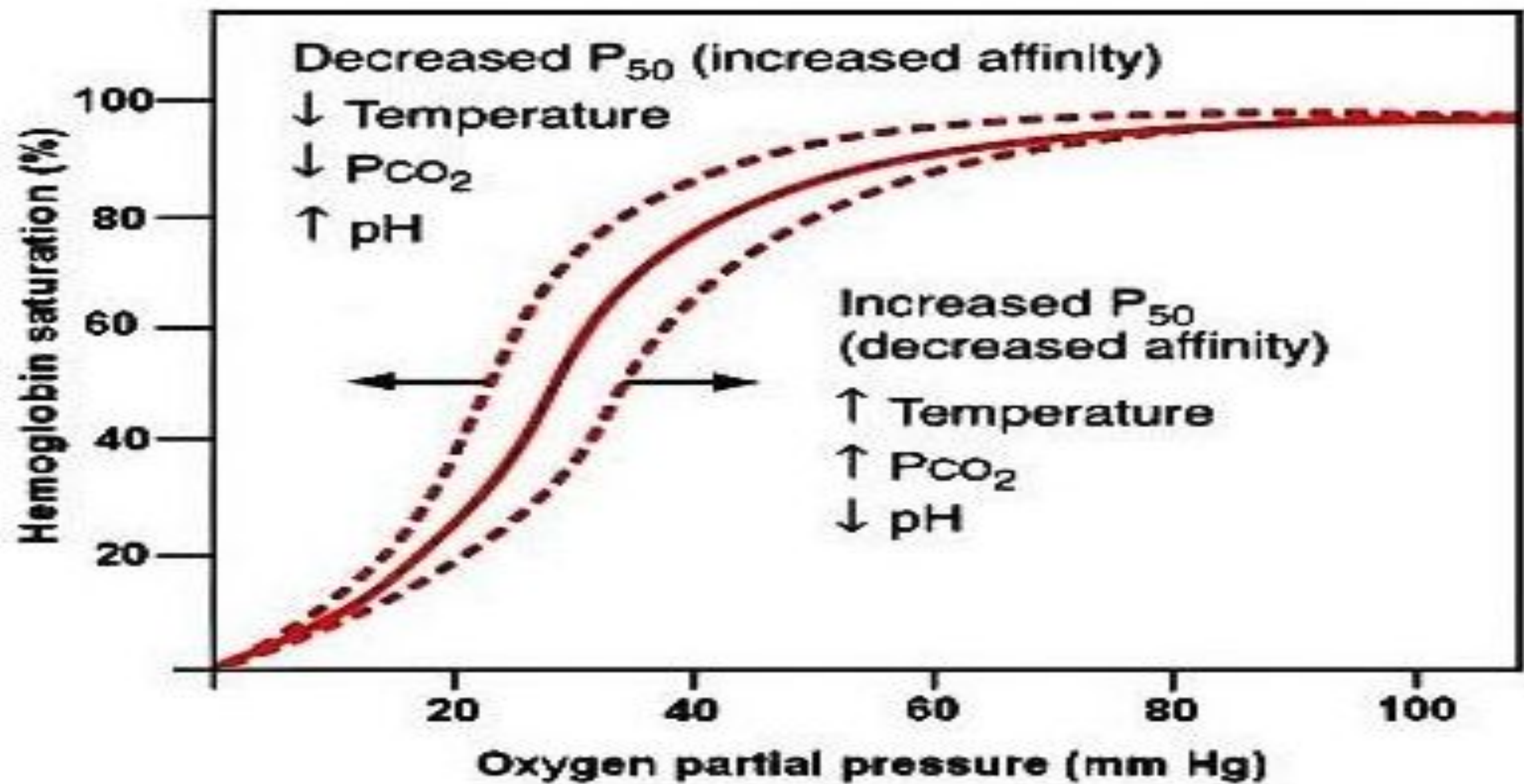
X-axis = partial pressure of oxygen(PO_2)

SHAPE:

Sigmoidal (s-shaped) curve which shows cooperative binding.

SIGNIFICANCE:

Enables efficient loading in lungs(high PO_2)and unloading in tissues(low PO_2)



OXYGEN DISSOCIATION CURVE

ZONES OF ODC(OXYGEN-HEMOGLOBIN DISSOCIATION CURVE)

- **PLATEAU ZONE**

High PO_2 (60-100 mmHg) occurs in lungs where even with large drops in PO_2 , saturation remains high.

- **STEEP ZONE**

Low PO_2 (<60 mmHg) occurs in tissues where small drop in PO_2 leads to large release of oxygen.

- **P_{50} VALUE**

The partial pressure of oxygen at which 50% of hemoglobin is saturated.

Normal P_{50} value = 27 mmHg

BOHR EFFECT

Bohr effect describes the reduction in hemoglobin's affinity for oxygen due to lower Ph (high acidity) or increased CO₂ concentration.

This phenomenon is important in oxygen delivery which causes a rightward shift in ODC forcing hemoglobin to release oxygen in actively metabolizing tissues.

Bohr effect can be described as:

The phenomenon where increased carbondioxide conc. & decreased pH (acidic environment) lower hemoglobin's oxygen-binding affinity facilitating increased oxygen release to tissues.

DESTRUCTION OF HAEMOGLOBIN

- After the lifespan of 120 days, the RBC is destroyed in reticuloendothelial system, particularly in spleen and the haemoglobin is released into plasma. The released haemoglobin is phagocytized immediately by macrophages in many parts of the body but specially by Kupffer cells of liver & macrophages of spleen and bone marrow. The haemoglobin is degraded & split into globin & heme. Globin is utilized for the resynthesis of haemoglobin & heme is degraded into *iron* & *porphyrin*. Iron is stored in the body as *ferritin* & *hemosiderin* which are reutilized for the synthesis of new haemoglobin. Porphyrin is converted into a green pigment called *biliverdin*. In humans most of the biliverdin is converted into a yellow pigment called *bilirubin*. Bilirubin and biliverdin together called the *bile pigments*.

HAEMOGLOBINOPATHIES

These are genetic disorders affecting either the structure of haemoglobin or the production of haemoglobin. They can result from:

- Abnormal globin structure
- Reduced globin synthesis

These disorders are among the most common inherited diseases worldwide.

ABNORMAL STRUCTURAL FORMS

HAEMOGLOBIN

- ***HAEMOGLOBIN S***

Haemoglobin S is found in *sickle cell anaemia*. In this type of haemoglobin, the alpha chains are normal while beta chains are abnormal. Amino-acid valine is substituted for glutamic acid at one point in each of two beta chains.

When this type of haemoglobin is exposed to low oxygen, it forms elongated crystals inside the RBCs. These crystals make it almost impossible for the cells to pass through small capillaries and the spiked ends of crystals are likely to rupture the cell membranes leading to sickle cell anaemia.

CONT..

- ***HAEMOGLOBIN C***

In haemoglobin C beta chains are abnormal in which amino-acid lysine is substituted for glutamic acid at one point of each chain. This type of haemoglobin is found in people with *Haemoglobin C disease* which is characterized by mild haemolytic anaemia and splenomegaly.

- ***HAEMOGLOBIN E***

In haemoglobin E also, beta chains are abnormal. Glutamic acid of beta chain is substituted by lysine. It is present in people with haemoglobin E disease characterized by mild hemolytic anaemia and splenomegaly.

CONT..

- ***HAEMOGLOBIN M***

It is abnormal type of haemoglobin and present in the form of methemoglobin. It occurs due to deficiency of cytochrome b5 reductase (methemoglobin reductase). Cytochrome b5 reductase keeps haemoglobin in normal (reduced) form. Deficiency of this enzyme results in methemoglobinemia (haemoglobin M disease). This disease in babies is called *blue baby syndrome*.

IRON METABOLISM

DISTRIBUTION OF IRON IN THE BODY

Total quantity of iron in the body is about 4g. Approximate distribution of iron in the body is as follows:

In haemoglobin : 65%

In muscles as myoglobin : 4%

As intra-cellular oxidative heme compounds : 1%

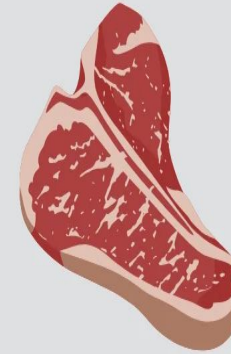
In the plasma as transferrin : 0.1%

Stored in reticuloendothelial system for later use : 25-30%

DIETARY IRON

Dietary iron is available in two forms:

1. **Heme iron:** present in fish, meat & chicken. Heme iron is absorbed easily from intestine.
2. **Non-heme iron:** present in vegetables, grains & cereals. Non-heme iron is not absorbed easily as heme iron.



VS



HEME IRON

EASILY ABSORBED

- From animal sources
- Absorption rate 15-35%
- Ready to use straight away
- No Anti-nutrients to compete with

NON-HEME IRON

POORLY ABSORBED

- From plant sources
- Absorption rate 1-15%
- Requires conversion before the body can use it
- Competes with compounds such as Phytates

ABSORPTION OF IRON FROM INTESTINE

Iron is absorbed from all parts of small intestine. It is absorbed through the intestinal cells (enterocytes) by pinocytosis and transported into the blood. Bile is essential for the absorption of iron because liver secretes moderate amounts of *apotransferrin* into the bile which flows through the bile duct into duodenum. Here the apotransferrin binds with the free iron and also with certain iron compounds like haemoglobin and myoglobin from meat. This combination is called *transferrin*. Then the transferrin molecule carrying its iron store is absorbed into the epithelial cells and later released into blood capillaries as plasma transferrin.

TRANSPORT OF IRON

When iron is absorbed from the small intestine, it immediately combines in the blood plasma with a beta globulin called *apotransferrin* to form *transferrin*, which is then transported in the plasma. The iron is loosely bound in the transferrin and consequently can be released to any tissue cell at any point in the body. In the cell cytoplasm, iron combines with a protein called *apoferritin* to form *ferritin*. Apoferritin has a molecular weight of about 460,000 and varying quantities of iron can combine in clusters of iron radicles. Therefore ferritin may contain only a small or large amount of iron.

STORAGE OF IRON

Excess iron in the blood is deposited mainly in the liver hepatocytes and less in the reticuloepithelial cells of bone marrow. In other cells also it is stored in small quantities. In the cytoplasm of the cell, iron is stored in large amount as *ferritin*. Small quantities are also stored in extremely insoluble form as *hemosiderin* . This is especially true when the total amount of iron in the body is more than the apoferritin can accommodate.

Hemosidrin collects in the cells in the form of large clusters that can be observed microscopically as large particles.

DAILY LOSS OF IRON

In males, about 1mg of iron is excreted everyday mainly into the feces. In females the amount of iron loss is very much high because of menstruation.

Each gram of haemoglobin contains 3.34mg of iron. Normally 100ml of blood contains 15g of haemoglobin and about 50mg of iron. So if 100ml of blood is lost from the body, there is a loss of about 50mg of iron.

In females, during every menstrual cycle about 50ml of blood is lost. So the iron content is always less in females as compared to males.

Iron is also lost during hemorrhage and blood donation. If for example 450ml of blood is donated about 225mg of iron is lost.

POST ASSESSMENT

Q1. Where does most haemoglobin synthesis occur?

- a) Liver
- b) Kidney
- c) Bone marrow
- d) Spleen

Q2. Which protein transport iron in the blood?

- a) Ferritin
- b) Transferrin
- c) Albumin
- d) Haemoglobin

CONT..

Q3. Which protein is the main intra-cellular iron storage protein?

- a) Transferrin
- b) Hemosiderin
- c) Ferritin
- d) Globin

Q4. In sickle cell anemia RBCs become rigid & crescent shaped under normal oxygen conditions(T/F)

The background is a soft watercolor wash in shades of yellow and orange, with a textured, painterly appearance. The colors are blended together, creating a warm and gentle gradient.

Thank You